Vision Mamba Technology Overview

Selective state-space modeling for image representations

Project: Privacy-Preserving Federated Learning Framework for Early Brain Tumor Classification Using Multi-Hospital MRI Images

Technical overview | Research and project-evaluation reference

1. What is Vision Mamba?

Vision Mamba adapts Mamba-style selective state-space sequence modeling to visual inputs. A 2D image or feature map is converted to a sequence of visual tokens; state-space operations model relationships across positions. Vision Mamba is a model family, not one single universally standardized implementation.

Key references: Gu and Dao, “Mamba: Linear-Time Sequence Modeling with Selective State Spaces” (2023); Zhu et al., “Vision Mamba: Efficient Visual Representation Learning with Bidirectional State Space Model” (2024).

2. Why use it for MRI?

- Convolutional backbones learn local patterns such as edges, textures and tissue boundaries.
- Mamba-style blocks can model dependencies between distant spatial tokens, complementing local convolutional features.
- State-space sequence modeling may be more memory-efficient than quadratic self-attention for long token sequences, depending on architecture and implementation.
- Potential benefit is a research hypothesis: improved cross-region context and non-IID robustness must be demonstrated, not assumed.

3. Architecture workflow

Stage	Typical tensor / function
MRI preprocessing	224 x 224 input, normalized with training-compatible statistics
EfficientNet-B0	Extract a spatial feature map B x C x H x W
Token projection	Flatten spatial positions into B x (H*W) x D tokens
Vision Mamba	Bidirectional or 2D-aware selective state-space token modeling
Pooling	Aggregate token features into a patient-image embedding
KAN classifier	Predict four diagnostic categories

4. Key concepts

State-space models maintain a latent state updated as sequence tokens are processed. Selective Mamba parameters depend on the input, allowing context-dependent information propagation. Bidirectional scanning helps visual tasks because images do not have a single natural left-to-right ordering.

Do not describe a generic convolution or attention layer as “Mamba.” A real implementation should use documented selective state-space operations, such as the official mamba-ssm package or a validated

Vision Mamba implementation.

5. Federated integration

- Each simulated hospital client trains the same hybrid architecture on its local dataset, using a label-aware objective when class vocabularies differ.
- FedAvg, FedProx and SCAFFOLD aggregate compatible shared model parameters; SCAFFOLD additionally maintains control variates.
- Compare global macro-F1, worst-client macro-F1, communication bytes, wall-clock time and GPU memory against an EfficientNet-B0-only baseline.
- For datasets assembled from overlapping public sources, deduplicate and audit provenance before treating results as independent-site evidence.

6. Strengths, tradeoffs and risks

Potential strength	Risk / validation need
Long-range spatial context	Image token ordering and scan direction affect representations
Efficient sequence modeling	CUDA extension and dependency compatibility can be difficult
Hybrid local + global features	More parameters and communication cost
New architectural comparison	No guaranteed performance improvement over CNN baseline

7. Colab implementation checklist

- Check the Colab GPU, torch and CUDA versions before installing mamba-ssm.
- Verify the model has a genuine Mamba block and print trainable parameters and intermediate tensor shapes.
- Run a forward/backward pass with synthetic tensors before downloading large datasets.
- Keep the same dataset splits and evaluation protocol for baseline versus hybrid.
- Record seeds, library versions, hardware and communication costs.

8. Viva questions and references

- Why Mamba over a transformer? It offers a different approach to long-range sequence modeling; practical efficiency must be benchmarked.
- What is the innovation? Evaluate the hybrid under heterogeneous-label federated MRI learning with measured robustness and calibration.

Gu, A., and Dao, T. (2023). Mamba: Linear-Time Sequence Modeling with Selective State Spaces. https://arxiv.org/abs/2312.00752

Zhu, L., et al. (2024). Vision Mamba: Efficient Visual Representation Learning with Bidirectional State Space Model. https://arxiv.org/abs/2401.09417